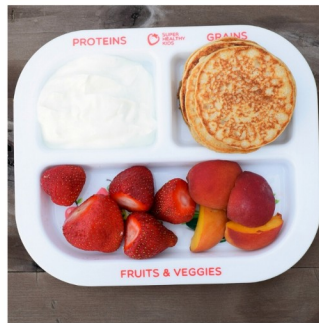
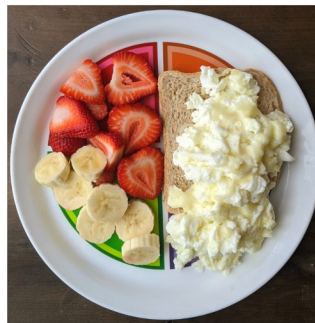


BREAKFAST

Lanthano

Aug 21st, 2025



Protein: Egg Whites
Fruit:
Strawberries/bananas
Grains: Toast

Protein: Greek Yogurt
Fruit: Strawberries/
peaches
Grains: Pancakes

Protein: Eggs
Veggies: Avocado
Fruit: Berries
Grains: Toast

BREAKFAST

healthbeet.org

EVERYONE KNOWS THAT YOU EAT THREE MEALS A DAY. Breakfast, lunch, and dinner. And we all know that breakfast is the most important meal of the day. It's a bad thing to skip it, it is even dangerous and harmful to your health.

This is what society has fed us. This is the norm. But let us look at whether this is actually true or not.

HISTORY

First off, three meals a day has never been the norm. Before the 1900s, people ate one or two meals a day. The typical was a midday meal (we call lunch) and dinner.

The shift started during the industrial revolution. As people began leaving their homes in the morning for work, they would take food with them or eat before they left. This is where breakfast started, but it mostly affected farmers and others around them^{A1}. But in the 1950s, as industrial foods began to appear, three meals a day was pushed through advertising, and eventually three meals became the norm we have today.

From what we know now, in Ancient Egypt common people had two meals a day. A light one at lunch time, and a heavier one later in the evening. Greece did the same. Rome actually did one heavy meal in the late afternoon. Later the elite in Rome started doing three meals a day. This is the first time in recorded history that three meals a day shows up.

BREAKFAST MARKETING

The phrase "breakfast is the most important meal of the day" was popularized in 1917 by the Grape Nut cereal brand^{A2}. This campaign was part of a larger plan by multiple cereal companies to make breakfast a crucial meal. They would often highlight the supposed health benefits of their products.

Edward Bernays, an extreme pioneer in public relations, conducted a campaign for Beech-Nut bacon in the 1920s. He persuaded at least 5,000 doctors to endorse bacon as a healthy breakfast option, leveraging their authority to influence public perception. This of course had nothing to do with actual science. Bernays used the doctors to promote Beech-Nut bacon and it wasn't based on anything^{A3}. He bought what he needed for a successful marketing campaign. And that is what got the idea of a heavy breakfast into the heads of Americans.

And now, today, we hold breakfast—the breakfast that Edward Bernays and cereal companies gave us—as a rule that can't be broken. We even see it as dangerous to break away from the idea and skip the meal.

"The notion that a daily morning meal is essential to health was heavily promoted by cereal companies in the 20th century, rather than based on strong scientific evidence."

Warren Belasco, food historian, *Meals to Come: A History of the Breakfast Industry*, 2008.

Even if someone breaks away from the cereal, eggs, and bacon breakfast. And starts eating a healthier meal every morning, they still don't question the concept of breakfast itself. They have the right idea, but often, they can't see outside of the box they start inside.

AROUND THE WORLD

Many countries around the world do not indulge themselves with breakfast. Some places it is the norm to skip it entirely, or maybe treat it like a light snack.

The reasons range from having no appetite in the morning after waking, to feeling weighed down or discomforted by an early meal.

In Spain, breakfast is just coffee and a small pastry. France and Italy do the same and some there see American breakfasts as unnecessary or excessive. Other places such as Germany or Scandinavia do have large breakfasts akin to American morning meals, but some of the younger generations have began moving away from such practices.

From what I have found, out of 195 countries, only around 25% (48 countries) engage in a large breakfast with the USA. 60% (around 117 countries) do a very light breakfast, and 15% (about 30 countries) do none at all^{B1}.

So, Americas idea of breakfast being the most important meal of the day, is largely an American idea. The majority of the world either does a light breakfast or none at all.

HEALTH

Skipping breakfast is very commonly considered unhealthy. But we shouldn't take this as fact without first examining it.

Breakfast is exactly what it says. It is a break of fasting. We are told that breakfast is the most important meal of the day, but there is a lot of evidence that indicates that fasting past breakfast can greatly benefit a person. Bionically we are even told that fasting is extremely beneficial.

Let's put this idea—breakfast—to the test. Let's see what the pros and cons are for skipping breakfast.

EATING BREAKFAST

Every morning when you eat breakfast, your body moves out of its fasting mode, and starts the daily process of breaking down the food you consume.

It is well known that a very important part of healing and health happens while you sleep^{C1}. And some believe that your body repairs so much of itself during the night because, in part, due to the fact that it is fasting.

When you wake up to a meal, your body immediately moves into dealing with that food. Nighttime healing and repairs stop and your body starts focusing on digestion.

FASTING AND AUTOPHAGY

If you wake up and instead skip the first meal of the day, your body remains in its fasting state. This allows it to keep putting energy toward other tasks, such as repairing the body and maintaining your health. This is when many people feel an abnormal level of alertness when compared to normally eating breakfast.

Autophagy is your body's "cellular cleanup" system. Damaged proteins and organelles are broken down and recycled. Research shows that fasting in the morning allows this process to continue into the day instead of stopping at breakfast.

Chaperone-Mediated Autophagy (CMA) begins around 30 minutes into starvation and remains elevated for several hours. If fasting continues beyond 10 hours, CMA reaches maximal activation by about 36 hours and stays high for up to 3 days, aiding in targeted protein recycling.^{C2}

So, while you can absolutely be fine with eating breakfast everyday. You might do even better long-term by skipping it. And this new research shows that a multi-day fast would have significant health benefits.

FASTING AND MMC

Migrating Motor Complex (MMC) is basically a wave-like pattern that moves through the digestive track and "resets" everything^{C3}. It cleans out the entire small intestine and resets it for when the next meal comes.

MMC only sets in after about four to five hours after a meal. The USDA and CDC don't give exact times, but they advise against irregular meal times and they emphasize the importance of waiting at least four hours before the next meal^{C4}.

While it is possible around meals and at night, skipping breakfast allows the gut to clean itself out much better and more thoroughly.

“Skipping breakfast can allow gut motility to complete its overnight cleansing cycles and helps maintain the natural rhythm of ghrelin and motilin, which regulate hunger and digestive housekeeping.”

Dr. Michael Camilleri, gastroenterologist, Nature Reviews Gastroenterology & Hepatology, 2021.

The average American has breakfast at around 8:12am, lunch at 12:00pm, and dinner at 6:15pm^{C5}. This gives around four hours between breakfast and lunch, and about six hours from lunch to dinner.

But this only allows MMC to work if there are no snacks eaten between meals. And unfortunately, 52% of Americans admit to regularly snacking between breakfast and lunch (between 9:30am and 11am); more than 70% say they regularly snack between lunch and dinner (between 2pm and 4:30pm; and around 67% claim to snack after dinner before bed (between 7:30pm and 9:30pm). Eating right before bed is very unhealthy, but what is really bad, is that some adults snack even later than that. Around 60% claim to have a late snack between 9pm and 11pm before bed^{C6}.

This data indicates that around 60% to 70% of Americans do not allow their bodies the fully complete an MMC cycle on a regular basis.

The MMC is the guts primary way of sweeping bacteria and undigested food out of the small intestine into the colon. And if this is prevented, as most Americans are likely dealing with, there is a list of side-effects that a person can go through^{C7}:

- Gas, bloating, abdominal discomfort.
- Feeling “heavy” or not fully digested after meals.
- Erratic bowel habits.
- Nutrient malabsorption.
- Indigestible debris, mucus, and cellular waste aren’t cleared as efficiently. This can lead to low-grade inflammation and sluggish motility.
- Motilin and ghrelin (which trigger MMC and hunger) rise during fasting, but frequent eating suppresses them. This can blunt natural hunger/satiety rhythms, possibly encouraging overeating.

Long term effects can hurt your metabolic health, and lead to chronic motility disorders. Constant MMC disruption is also linked to obesity and weight gain^{C8}.

Looking at the list of side-effects for disrupting MMC, we see a significant increase since the year 2000. Since then, obesity overall has increased from 30.5% to about 42%. That's more than an 11 point increase. Severe obesity has roughly doubled, going from 4.7% to 9.2% since the year 2000. Bloating, gas, and abdominal discomfort has more than doubled in the last 25 years. Rising from 19% in 2000, to 44% to 73% in 2025^{C9}.

SKIPPING BREAKFAST

So now there is a question. Should we skip breakfast? Yes and no. Some people should, and some people shouldn't. The benefits and drawbacks aren't universal across the board. There are a lot of benefits, and there are some possible side-effects. It depends on the person.

Let's look at the pros and cons of skipping breakfast.

PROS

Contrary to popular beliefs, there are many health benefits for skipping breakfast mealtime. This isn't universal, and many people will talk about how horrible they feel when you miss out of breakfast. But there is a large part of the population that sees great benefits from skipping the meal regularly.

Here's a list of some of those benefits^{D1}.

- Skipping breakfast naturally shortens your daily eating window, which can help with calorie control and sometimes weight loss.
- Some people experience better digestion when they skip breakfast. This would be due to allowing the digestive system to rest and recover from the previous day's meals.
- Fewer eating occasions can reduce blood sugar spikes, which may benefit insulin sensitivity and metabolic health.
- Some studies show intermittent fasting helps people with prediabetes or at risk of type 2 diabetes.
- After an overnight fast, skipping breakfast extends the fasting state, encouraging the body to use stored fat for energy.
- Many people report feeling more alert when fasting, since insulin and blood sugar fluctuations are minimized.
- Autophagy is a natural cellular process where the body removes damaged cells and recycles components for energy. Intermittent fasting, achieved through skipping breakfast, has been shown to enhance autophagy.
- Studies suggest that intermittent fasting, achieved through skipping breakfast, may lead to reduced inflammation in the body.
- Some evidence suggests fasting can increase levels of norepinephrine, which boosts focus and energy.

CONS

Those who warn against the idea of skipping breakfast will point to these side-effects^{D2}.

- For some, skipping breakfast can impair concentration, memory, and reaction times, particularly in the late morning. Athletes or physically demanding jobs may see a dip in performance without morning fuel.

- A few studies suggest skipping breakfast may possibly increase insulin resistance, blood pressure, and cholesterol levels in certain people.
- Some warn that skipping breakfast could possibly be bad for your heart. The thought is that because your body thinks it's starving, it has an adrenaline dump. And after a while this starts hurting your heart and can lead to heart attacks.

As you will see in the next section, the heart attack claims are wildly debated and very likely not truly connected with skipping breakfast. But skipping breakfast for some just doesn't work. Some people just can't deal without the first meal of the day. And that's not a bad thing.

"Whether breakfast is beneficial depends on the person's lifestyle, activity level, and total caloric intake. Some people perform better skipping it; others feel lethargic."

Dr. Tim Spector, epidemiologist and author of *The Diet Myth*, 2015.

HEART HEALTH

A 2019 study pooling four prospective cohort studies (over 199,634 adults) found that those who regularly skipped breakfast had a 21% higher risk of cardiovascular disease and a 32% higher risk of all-cause mortality compared to those who regularly ate breakfast^{D3}. This research is untrustworthy due to the fact that definitions varied throughout the study and residual confounding remained possible.

Skipping breakfast is bad for you. A study from Harvard/Brigham in 2013 found that men that skip breakfast regularly have a 27% higher chance of having a heart attack when compared to those that eat regularly.

This is heavily debated though.

BMJ Nutrition, Prevention & Health in 2020 to 2022, found that skipping breakfast alone does not increase the chances of heart attacks^{D4}. It found late night eating is likely what increases this chance and not skipping breakfast alone.

American Heart Association remains neutral on the subject, saying that more research need done before decisive conclusions can be reached one way or the other^{D5}. Experts at the Mayo Clinic point out that the benefits from fasting remain largely unverified. And a recent article from Verywell Health explains that skipping breakfast could be dangerous to heart health or could be very beneficial.

There is even research out there that indicates that regular breakfast eaters may have a higher chance of cardiovascular death^{D6}. But, this needs more research before certain claims can be made.

Overall, it is unwise to jump on articles and research like this, and claim skipping breakfast is harmful, due to the arguable nature of these studies.

While several studies show associations between skipping breakfast and increased cardiovascular risk, many of these findings can be explained by other factors—like poor lifestyle, erratic meal timing, or metabolic imbalances—rather than the act of skipping breakfast itself. Similarly, intermittent fasting may offer metabolic benefits, but its impact on long-term heart health, especially across different protocols, remains inconclusive.^{D7}

While this area remains open for study and research, the current research leans toward skipping breakfast as beneficial. Long term studies are lacking, but the short term appears to be promising.

"The idea that breakfast is the most important meal of the day is largely cultural and marketing-driven. For many adults, skipping breakfast does not harm metabolism or cognitive function."

Dr. Satchin Panda, circadian biology researcher, *Cell Metabolism*, 2019.

BREAKFAST WITHDRAWALS

Some people claim that they stopped eating breakfast as a test and it wasn't even a day before they had all kinds of horrible things happening to them. The two biggest ones usually reported are: hunger, and irritability or mood swings.

This is breakfast withdrawal (however not a clinical term). But that's what it is. Any habit that someone tries to break can come with withdrawal symptoms of its own kind, and breakfast is not different in this aspect.

The biggest thing that people notice while trying to train their body without breakfast is hunger. But here is a more complete list of side-effects people experience^{D8}:

- Headaches: from the temporary drop in blood sugar.
- Irritability or mood swings: low morning glucose can affect mood.
- Fatigue or brain fog: energy dips can feel pronounced if your body expects breakfast.
- Hunger pangs or nausea: the stomach may “remember” its usual routine.

These are most prevalent in the first three days. After about four to seven days, the body starts adjusting and the symptoms begin to subside. Symptoms are usually gone by one or two weeks.

So, just because you or someone you know skipped breakfast for a couple days, and thought it was terrible, doesn't mean it actually is. That would be like trying to stop smoking, feeling how bad it is and assuming that it never gets better. It does. There's just a barrier that has to be pushed past.

EATING LIFE AWAY

There is a scientific connection between eating more and dying younger.

Moderate caloric restriction (10% to 40% less than you normally need) without malnutrition is associated with longer lifespan and better health in many species, with promising signs in humans. 10% to 40% is like going from 2,000 calories a day, to about 1,200 to 1,800 calories per day.

Caloric restriction (CR) is very promising in animals of every kind.

Rodents, worms, flies, and yeast: Across dozens of experiments, 20–40% reduction in calories (without malnutrition) consistently extends lifespan, sometimes by 30–50%.

Monkeys: Two long-term rhesus monkey studies (UW–Madison and NIA) found mixed results, but both showed that CR reduced age-related diseases like diabetes, cancer, and heart disease. One study also showed improved survival.

Animal trials show a great increase in longevity. And human trials show great promise too.

The CALERIE Phase 2 trial, a 24-month randomized controlled study, assigned healthy, non-obese adults to either a ~25% calorie-reduction intervention or an ad libitum (normal diet) control. Participants achieved an average of around 11–12% calorie restriction, maintaining about a 10% weight loss over two years. This CR group showed improvements in metabolic markers like insulin sensitivity, cholesterol, reduced blood-insulin and inflammation markers—without negative effects on quality of life.^{E1}

A follow-up explored the impact of CR on DNA methylation-based aging measures (the so-called epigenetic “clocks”). CR notably slowed the pace of aging by about 2–3%, as measured by DunedinPACE, suggesting a potential 10–15% reduction in mortality risk—a magnitude comparable to the known benefits of quitting smoking.^{E2}

Basically, by eating less, you can noticeably increase your own lifespan. Inversely, this means that by eating more, you're taking time off your clock.

So, while obviously going through modern life eating three meals a day is totally normal, skipping breakfast might have benefits that could add years to your life.

"To lengthen thy life, lessen thy meals."

Benjamin Franklin

This isn't a warning, it is wisdom, wisdom that was given long before we had all this modern science and research that shows us that Benjamin Franklin and others like him were right.

CALORIC RESTRICTION

By how much would CR add to your lifespan? In mice, a mouse that normally lives two years, gets close to three.

And while animal trials are not easily translated to human patients and trials, we still have a fair amount to work with. When it comes down to direct numbers, CR might add anywhere between two and 13 years of life in humans^{E3}. More conservative skeptical numbers say CR only adds around two or three years to human lifespans^{E4}.

One place that could perhaps showcase CR is Okinawa in Japan. The traditional Okinawan diet is around 1,800^{E5} calories a day, compared to 2,500 in mainland Japan and even higher in America.

What is special about the Okinawan people, is that they have the world's highest lifespans on record (non biblical). The average Okinawan life expectancy is 86 years for women and 78 years for men^{E6}. The world average for women is 76 years and for men is about 71^{E7}. That's an average of 10 extra years for women, and seven years for men.

Okinawa also has up to five times the number of people living past the age 100 than most other countries. The Okinawan people are a gem for the scientific community, and are currently being researched. While there is much more that needs examining, it is possible that CR has at least a small role to play in the increased longevity.

WHEN TO EAT

One thing that can have a large impact on your health in a bad way is eating late at night. In fact, this is one of the factors that led the Harvard/Brigham study in 2013 toward the idea that skipping breakfast was unhealthy. It wasn't the skipping of breakfast that was harmful, but eating late at night that caused the increased chance of heart attacks.

The body's metabolism follows a day and night cycle. Eating late can throw off the body's circadian rhythm. This can lead to poorer glucose control and altered hormone release (like insulin, cortisol, melatonin). Over time, this can increase risk for metabolic syndrome, obesity, and diabetes.

That is what the Harvard/Brigham study of 2013 said skipping breakfast caused. But in reality, this is caused by eating late at night, not skipping breakfast.

Along with that, eating late at night comes with other drawbacks and issues^{F1}:

- Eating close to bedtime can cause acid reflux, indigestion, or discomfort while lying down. This may also disrupt sleep architecture, making sleep lighter or less restorative.

- Calories eaten late are more likely to be stored as fat because insulin sensitivity is lower at night.
- Studies suggest people who eat late may consume more total calories and make poorer food choices (snacking, high sugar/fat).
- Some observational studies link late-night eating with higher blood pressure, worse lipid profiles, and greater heart disease risk.
- Eating late in the evening (especially carb-heavy meals) raises nighttime glucose levels.
- In people with diabetes or prediabetes, this can worsen insulin resistance and morning blood sugar readings.
- Late-night eating has been associated with higher cortisol levels and interference with natural fasting-related repair processes (like autophagy).

Regardless of whether you skip breakfast or not, it is recommended that you finish your last meal of the day at least two or three hours before you go to sleep. Experts in the field recommend eating between 5pm and 7pm.

While doctors and dietitians acknowledge that schedules and things outside of our control all influence dinnertime, they seem to agree on an ideal window of time. “Eating between 5 to 7 p.m. would be ideal,” says Dana Cohen, M.D., integrative medical doctor and co-author of Quench.^{F2}

In the end, research does not indicate that skipping breakfast is unhealthy, but that eating late is. Whether or not you eat breakfast, keep in mind that it is wise to eat earlier in the evening.

WITH BREAKFAST

With breakfast it is recommend that you space your meals out by at least four or five hours. This puts the meals at about this time for someone who is up at around 6am.

Breakfast: 7am to 9am

Lunch: 12pm to 1:30pm

Dinner: 5:30pm to 7:30pm

It is recommend that you eat breakfast (if indeed you are doing so) about one or two hours after you wake up. This is when your cortisol is naturally high, and this helps mobilize energy. Eating soon after you wake up can help stabilize blood sugar and set circadian cues.

WITHOUT BREAKFAST

Without breakfast, the day is mostly the same.

Breakfast: skipped, allowing fasting to continue

Lunch: 12pm to 1pm

Dinner: 5pm to 7pm

This allows for four to seven hours between meals, and 17 to 20 hours of fasting. The extra time allotted to fasting allows for a huge rise in Autophagy and cell regeneration, and allows multiple cycles of the MMC to complete. And as long as the remaining two meals aren't larger than they normally are, you can also be cutting down on calorie intake. Cutting out a normal breakfast could easily result in

about 15% to 25% calorie-reduction^{F3}. However, if meals become larger later in the day (which is not unreasonable), the calorie-reduction would be reduced to only 10% or less.

"Extending the overnight fast by skipping breakfast can improve insulin sensitivity, reduce body weight, and support metabolic flexibility without requiring calorie counting."

Dr. Krista Varady, intermittent fasting researcher, The New England Journal of Medicine, 2019.

MY EXPERIENCE

I usually don't include sections that focus on me, but since I have a little experience here, I guess it wouldn't hurt to explain myself.

I had eaten three meals a day for about 17 years. I thought nothing of it. When I got my first job (which required me leaving home at around 6am), I started skipping breakfasts. What I immediately noticed was the increased alertness.

This was a desk job and I can't say it was the kindest to me. I left that job for construction work, concrete work to be exact. And the habit of skipping breakfast continued. Coworkers told me I wouldn't make it without the morning energy, and my parents tried to get me to make myself breakfast each morning. But, because I still had to be up early, and because I am too lazy, I continued to skip it.

And contrary to what I was told, I did fine. I wasn't crushed by the work, and I had plenty of energy each morning.

Occasionally I would cook myself something in the morning, or grab a meal from a gas station before work. But the food (no matter where it came from) made me feel sluggish for the rest of the day. It wasn't supper horrible, painful, or uncomfortable. But it was enough that I decided to just not eat breakfast.

Fast forward to my current job (washing windows, pressure washing, cleaning gutters, etc.), I still rarely eat breakfast. When I do, I still notice the little lack of mental clarity. I'm sometimes hungry enough that I don't care. But for the most part, skipping breakfast feels great. I am not starving or suffering because of it, and I really believe it to be beneficial to me.

I can't say whether or not I had more energy than when I ate breakfast, but with the mental clearness, it definitely felt like it.

And this all happened before I knew anything about the benefits of skipping breakfast. I'm 19 now and just starting to research why I feel so much better without breakfast. I didn't research this and then test it. I tested skipping breakfast without knowing I was, and then dug into the research of why it felt better than eating.

CONCLUSION

There are many benefits that come with skipping breakfast. Some people experience higher energy levels, this group would include myself. Others experience lower energy, I know people that fit in that group. In the end, this is not a one size fits all, and we shouldn't treat it that way.

And unless more research is brought to my attention, I see no reason for as to why skipping breakfast is a universal bad thing to do. And I certainly don't see why it's bad for me.

If skipping breakfast makes you grumpy, groggy, or you just don't want to do it. Then go ahead and eat, I'm not saying you have to skip it. I'm just saying that there are beneficial reasons for avoiding breakfast.

This is not me talking bad about breakfast, this isn't me being all negative on your parade. This was just me researching why I felt better without that first meal.

SOURCES

Video sources in this paper are available for download from my ProtonDrive at:

<https://drive.proton.me/urls/0KT6J9V40W#ak5ghv46ESsy>

* Indicates a topic or source that was uncovered with the help of an AI model (ChatGPT, Grok, etc.). The facts and sources themselves are independently studied and verified by the author. No AI model is taken at it's word.

Email TheOregonTrail@protonmail.com if you would like to ask a question, found an error throughout this paper, or have more evidence you would like to present to me (regarding either side of the debate).

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“Three meals a day, especially with a substantial morning meal, was never a universal human practice; it became standard in the United States largely through social engineering and commercial promotion.”

Warren Belasco, Food: The Key Concepts, 2008.

Never take something that culture gives you at face value. God set things in perfect order once. Man has since messed it up, and if we ever want things to be better, we only need move back toward God and what He originally created.

To God be the glory forever and ever.

**Be not among winebibbers; among riotous eaters of flesh:
For the drunkard and the glutton shall come to poverty: and drowsiness
shall clothe a man with rags.**

Proverbs 23:20-21